

CM Yang

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Summary

- 8 years of experience in software development and system design related to data engineering
- 2-3 years of experience in deep learning for computer vision, specializing in segmentation and tracking
- Familiar with development in Python and Golang
- Familiar with CI/CD and Kubernetes operations
- Experienced in building data pipelines and data warehouses in both GCP and AWS environments
- Experienced in building data teams from scratch on two separate occasions
- Passionate about all aspects of data, including data science research, data infrastructure, data governance, and data analysis

Technical Skills

- **Languages:** Python, Go
- **Tools & Infrastructures:** Kubernetes, Airflow, Argo Workflow, BigQuery, Redshift, RDS, Kinesis
- **Frameworks & Skills:** System Monitoring, Data Warehousing, Data Pipeline Design, Pub/Sub

Education

National Chi-Nan University (NCNU)

Department of Computer Science and Information Engineering, Master Degree

Nantou, TW

Sep. 2015 - Jun. 2016

Memorial University of Newfoundland (MUN)

Department of Computer Science, Bachelor Exchange Student

Nantou, TW

Sep. 2014 - Jun. 2015

National Chi-Nan University (NCNU)

Department of Computer Science and Information Engineering, Bachelor Degree

Nantou, TW

Sep. 2011 - Jun. 2015

Experience

Kronos Research

Software Engineer II, Data Platform Team

Taipei, TW

Nov 2021 - Present

- As the Product Owner of the High-Frequency Trading data collector, I designed and implemented the data flow, collaborated with cross-functional teams to ensure data quality, and optimized operational costs.
 - a. [Infrastructure] Migrated the logging system to an event-based architecture and redesigned the entire ecosystem, achieving an 85% cost reduction.
 - b. [Operation] Implemented a unified data quality interface via API for traders, seamlessly integrating it into the trading desk pipeline. This integration reduced redundant simulation jobs and eliminated the risk of dataset pollution, thereby enhancing overall model performance.
 - c. [Operation] Defined quality metrics by analyzing data to identify issues attributed to a limited set of tags, and monitored Grafana to address the root cause effectively.

- d. [Infrastructure] Redesigned and performance-tuned the dataset statistics job using the Argo pipeline to orchestrate 70,000 jobs. Optimized processing time from 21 hours to 3 hours using CGo.
- e. [Infrastructure] Collaborated with the HPC team to design a universal resource interface via API, enabling users to perform CRUD operations on machine resources. This interface leveraged the endpoint to automatically scale the project's machine resources during the planning stage.
- As the data platform team members, we are dedicated to fetching diverse data and designing various interfaces to deliver data to help traders and save costs simultaneously.
 - a. [Infrastructure] Developed a real-time pipeline for monitoring PNL in risk control, utilizing AWS Kinesis Data Streams to ensure data landing within one minute.
 - b. [Infrastructure] Implemented a REST API service in Go as a universal data provider interface, leveraging Argo for worker pools on EKS. Designed the API to be truly stateless.

WeMo Scotter Corp.

Data Scientist, Data Team

Taipei, TW

Sep 2019 - Oct 2021

- [Infrastructure] Enhanced the robustness of data pipelines and decreased time-to-production by transitioning the strategy from ETL to ELT. Utilized Apache Airflow with Jinja templates to establish standardized pipeline patterns and leveraged BigQuery as the data warehouse.
- [Infrastructure] Leveraged the Metabase for business reporting and hosted internal tutorials for non-technical employees to enhance their analytical capabilities, significantly reducing their dependency on the data team.
- [Analysis] Conducted geospatial analysis on trajectory datasets using Python and PostgreSQL, applied HDBSCAN for clustering, and identified hotspots using Voronoi diagrams. Provided insights to optimize resource relocation and dispatch strategies.
- [Analysis] Analyzed sensor hub data from free-float e-scooters to identify areas with low network quality. Collaborated with the top two telecommunications providers to enhance network performance, resulting in a mutually beneficial strategy.
- [Analysis] Analyzed user segments by examining user session data and AAARR funnel metrics from Firebase and BigQuery.

Academia Sinica, Institute of Information Science

Research Assistant and RDSS

Taipei, TW

Aug. 2016 - Aug. 2019

- [Analysis] Utilized web crawling and IP tracking techniques to analyze cyber-warrior's behavior on Taiwan's largest forum, identifying relationships between accounts and trending topics. Presented the findings to the mayor
- [Research] Supported a biology research paper published in Nature Communications by developing preprocessing tools with PyQt5, training a Mask-RCNN model for segmentation, and analyzing features using t-SNE.
 - [Paper - Artificial intelligence reveals environmental constraints on colour diversity in insects](#)
 - [Slides - Moth Analysis](#)
- [Research] Assisted a biology research experiment by extracting trajectories from six years of video footage in two phases. Employed the ReInspect Model for detection, ensembled ResNet models for classification, and applied Hungarian Matching with SORT to enhance robustness.
 - [Slides - Beetle Tracking](#)
 - [Github - Beetle Tracking](#)

- Conduct the data analysis workshop at "Data Science at Taiwan" Conference and "Taiwan AI Academic" organization covering topics from web crawling to textual analysis